

Telco Customer Churn Analysis

Summary and Recommendation

This analysis was conducted on the IBM Telco Customer Churn dataset containing 7,043 customer records and 21 variables related to demographics, account information, subscribed services, billing methods, and churn status. The objective was to identify the major drivers influencing customer attrition and uncover actionable insights to improve retention.

Overall Churn Rate

The analysis reveals that **26.54% of customers have churned**, meaning roughly one in four customers discontinued their service. While the majority of customers remain active, this churn rate indicates a significant opportunity for retention-focused strategies.

Key Insights

1. Gender Has Minimal Impact on Churn

The churn distribution across male and female customers is nearly identical, indicating that **gender is not a significant predictor of customer churn**.

2. Senior Citizens Show Higher Churn Propensity

Although senior citizens make up a relatively small portion of the customer base, they exhibit a **higher proportion of churn compared to non-senior customers**. This suggests that older customers may face challenges such as pricing sensitivity, service complexity, or inadequate support.

3. Tenure Strongly Influences Retention

The tenure analysis shows a clear pattern:

- Customers with **short tenure (0–12 months)** are far more likely to churn.
- Customers with **longer tenure** tend to remain loyal.

This indicates that the first year is the most critical period for customer retention, making onboarding and early engagement essential.

4. Contract Type Is the Strongest Predictor of Churn

Contract length has a substantial impact on customer behavior:

- Customers on **Month-to-Month contracts** exhibit the highest churn rates.
- Customers on **One-Year and Two-Year contracts** are significantly more likely to stay.

This demonstrates that long-term commitments greatly enhance customer loyalty and reduce attrition.

5. Service Adoption Improves Retention

Customers who subscribe to value-added services tend to remain with the company longer. Specifically, lower churn is observed among customers using:

- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming Services
- DSL Internet Service

In contrast, customers without these services show noticeably higher churn, suggesting that service bundles increase perceived value and customer stickiness.

6. Payment Method Significantly Affects Churn

Among all payment methods, customers using **Electronic Check** have the highest churn rates. Customers paying via automatic methods such as bank transfers or credit cards demonstrate better retention, indicating that billing convenience and automation may improve loyalty.

Business Recommendations

Based on the findings, the following strategies are recommended:

Customer Retention Initiatives

- Focus on customers within their first 12 months.
- Develop onboarding campaigns and proactive support programs.

Contract Conversion Programs

- Offer incentives to convert Month-to-Month customers into annual or multi-year contracts.

Service Bundling

- Promote security, backup, and technical support packages to increase customer dependence on the platform.

Payment Method Optimization

- Encourage customers using Electronic Check to switch to automatic recurring payments.

Senior Citizen Support

- Introduce dedicated support, simplified plans, and targeted outreach for senior customers.
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Conclusion

The analysis identifies **contract type, tenure, subscribed services, and payment method** as the most influential factors affecting customer churn. Customers who are new, on flexible monthly contracts, lacking additional support services, and paying by Electronic Check are at the greatest risk of leaving.

By focusing on these high-risk segments and implementing targeted retention strategies, the company can significantly reduce churn, improve customer satisfaction, and enhance long-term revenue stability